

INSuRE Fall 2025 Final Report

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Enhancing Multimedia Forensics and Authentication Analysis through Enhanced Explainability and Summarization of Results

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Contents

1	Executive Summary	3
2	Introduction	4
2.1	Top Level Research Project Description	4
2.2	Purpose Statement	4
2.3	Motivation	5
3	Literature Review	5
4	Methods and Procedures	7
4.1	Methods Used	7
4.2	Dataset Enrichment	7
4.2.1	VideoSham Dataset	7
4.2.2	Dataset Statistics	8
4.2.3	Annotation and Explainability Pipeline	9
4.2.4	Prompt Structure	9
4.2.5	Human–Model Refinement Loop	10
4.3	Model Setup	11
4.4	Procedures	13
4.5	Schedule	14
4.6	Deliverables	14
4.7	Limitations and Delimitations	15
5	Findings	15
5.1	Findings Overview	15
5.2	Detailed Findings	16
6	Issues and Challenges	19
7	Conclusions and Recommendations	21
8	Future Work	22
9	References / Bibliography	23
10	Team Biographies / Tasking	24

A	Acronyms and Terms	26
A.1	Acronyms	26
A.2	Keywords	27
A.3	Terminology	28

1 Executive Summary

The rapid progress of generative artificial intelligence has made it increasingly easy to create realistic but deceptive multimedia content, intensifying concerns around authenticity, trust, and disinformation. While existing forensic detectors can often identify manipulated images or videos, they typically produce only binary labels or confidence scores, providing limited insight into *where*, *how*, or *why* a piece of media has been manipulated. This lack of explainability increases the burden on human analysts and undermines trust in forensic decision-making. Although recent efforts such as the ExDDV [2] and VidGuard-R1 [7] have introduced benchmarks for explainable deepfake detection, they primarily focus on face-centric or fully generated content and provide limited coverage of general video manipulations and audio-visual inconsistencies.

In this project, we investigate *general-manipulation explainable multimedia forensics*, moving beyond face-based deepfakes. We curate an explainable forensics dataset based on VideoSham [5] that covers six common manipulation types, including entity addition and removal, background or color changes, text replacement, inter-frame manipulations, and audio replacement. Each manipulated video is paired with a structured natural-language explanation that describes the suspected manipulation, relevant visual, temporal, or audio artifacts, and a concise forensic summary.

Building on this dataset, we evaluate an open-source omni-modality large language model, Qwen-2.5-Omni-7B [9], for explanation generation under both zero-shot and fine-tuned settings. Using a carefully designed prompt and regularization strategy, we benchmark model performance with automatic metrics (BLEU-1 [6] and semantic cosine similarity) as well as human judgment of explanation consistency. Experimental results show that LoRA-based [3] fine-tuning substantially improves explanation quality compared to the zero-shot baseline, while also revealing failure cases, particularly for subtle frame-level manipulations.

Overall, this work introduces a practical task setting and dataset for general-manipulation explainable forensics and provides empirical evidence on the strengths and limitations of current omni-modality models. The findings highlight both the potential of large multimodal models for forensic explanation generation and the remaining challenges in achieving reliable, human-aligned explainability for real-world multimedia analysis.

2 Introduction

2.1 Top Level Research Project Description

This project addresses the critical need for explainability in multimedia forensics by focusing on how manipulated videos can be analyzed and explained in a human-interpretable manner. While existing manipulation detectors are often able to flag suspicious artifacts in images, audio, or video, they typically output only binary labels or confidence scores, offering limited insight into the underlying forensic evidence. Such opaque outputs slow investigations, increase the cognitive burden on analysts, and undermine trust in forensic conclusions.

Recent datasets such as ExDDV [2] and VidGuard-R1 [7] have taken important steps toward explainable deepfake detection by introducing benchmarks with textual explanations. However, existing benchmarks primarily focus on face-centric or fully generated content and provide limited coverage of general video manipulations and audio-visual inconsistencies. As a result, they do not adequately reflect the diverse manipulation patterns encountered in real-world multimedia forensics scenarios.

In this work, we move beyond face-based deepfakes and study *general-manipulation explainable multimedia forensics*. We curate an explainable forensics dataset based on VideoSham [5] that covers six common manipulation types, including entity addition and removal, background or color changes, text replacement, inter-frame manipulations, and audio replacement. Each manipulated video is paired with a structured natural-language explanation describing what appears manipulated, where or when anomalies occur, and which visual, temporal, or audio artifacts support the forensic judgment. Using this dataset, we evaluate an open-source omni-modality large language model for forensic explanation generation under both zero-shot and fine-tuned settings. Through systematic benchmarking with automatic metrics and human evaluation, this project provides empirical insights into the strengths and limitations of current multimodal models for explainable multimedia forensics.

2.2 Purpose Statement

The purpose of this project is to investigate whether large omni-modality language models can generate reliable, evidence-based natural-language ex-

planations for general video manipulations beyond face-centric deepfakes. By introducing a practical task setting, curating an explainable dataset, and evaluating zero-shot and fine-tuned models, this work aims to better understand how explainable multimedia forensics systems can support human analysts in real-world decision-making.

2.3 Motivation

As manipulated multimedia becomes increasingly prevalent in misinformation campaigns, fraud, and legal disputes, the ability to verify the authenticity of digital media is critical. Although modern forensic detectors can achieve high accuracy on benchmark datasets, they often provide only raw scores or binary outputs, offering little explanation of the evidence behind their decisions. This lack of transparency limits their usefulness in high-stakes forensic workflows, where analysts must understand not only whether content is manipulated, but also why.

Moreover, detectors that perform well under controlled benchmark conditions frequently struggle when applied to compressed, noisy, or previously unseen real-world content, raising concerns about robustness and generalization. These challenges are compounded by the lack of standardized datasets and evaluation protocols for explainable multimedia forensics, making it difficult to systematically assess and compare explanation quality across methods.

While prior efforts such as ExDDV [2] and VidGuard-R1 [7] represent important progress toward explainable deepfake detection, their focus remains limited in both manipulation diversity and modality coverage. In particular, general video manipulations and audio-visual inconsistencies are underexplored. Motivated by these gaps, this project seeks to broaden the scope of explainable forensics and to empirically evaluate how well current omni-modality models can reason about and explain diverse manipulation types in realistic multimedia scenarios.

3 Literature Review

The field of multimedia forensics has made substantial progress in the development of deepfake and manipulation detection methods, largely driven by advances in generative artificial intelligence (AI). Despite impressive detec-

tion accuracy on benchmark datasets, a persistent limitation remains: most forensic systems provide little to no explanation for their predictions. As a result, analysts are often left with binary decisions or confidence scores without sufficient context, evidence, or reasoning to support forensic conclusions.

Recent studies have emphasized the importance of explainability from a user-centered perspective. Wu et al. [8] show that intelligence analysts require detection tools that are not only accurate but also interpretable and trustworthy. Their findings highlight the need for explanations that convey why a piece of media is flagged as manipulated, rather than simply reporting a classification outcome. Similarly, Abbas et al. [1] provide a comprehensive survey of deepfake detection techniques and observe that the majority of existing approaches prioritize detection accuracy, with explainability often treated as a secondary or optional component. While some methods visualize intermediate features or saliency maps, these outputs are rarely sufficient for human analysts to form reliable forensic judgments.

Motivated by the broader field of Explainable AI (XAI), several efforts have begun to incorporate explanation mechanisms into multimedia forensics. Hondru et al. [2] introduce ExDDV, the first dataset explicitly designed for explainable deepfake detection in video. ExDDV includes manually annotated textual descriptions and spatial click annotations that indicate visible artifacts, enabling the evaluation of vision-language models for artifact localization and description. While this work represents a foundational step toward explainable forensics, its scope is primarily limited to face-centric or fully generated content, and the resulting explanations focus largely on localized artifacts rather than holistic forensic reasoning across visual, temporal, and audio modalities.

Beyond explainability, the evaluation and generalization of deepfake detectors remains an open challenge. Le et al. [4] present a systematization and benchmarking of deepfake detection methods, demonstrating that many detectors fail to generalize to real-world manipulations despite strong benchmark performance. However, their analysis focuses on detection accuracy and robustness, without explicitly addressing explanation quality or human interpretability. This further underscores the lack of standardized datasets and evaluation protocols for explainable multimedia forensics.

In summary, prior work has established strong foundations for detecting manipulated media and has begun to explore explainability through artifact-level analysis and visualization. Nevertheless, there remains a clear gap in research on explainable forensics for *general video manipulations*, particularly

those involving diverse visual edits and audio-visual inconsistencies. Existing methods rarely provide structured, natural-language explanations that align with how human analysts reason about manipulated content. This gap motivates the need for task settings, datasets, and evaluations that explicitly target general-manipulation explainable multimedia forensics, as explored in this work.

4 Methods and Procedures

4.1 Methods Used

Our methodology follows a data-driven and empirical approach to investigate explainable multimedia forensics for general video manipulations. The core methods include dataset curation and annotation, prompt-based explanation generation, and systematic evaluation of omni-modality large language models under zero-shot and fine-tuned settings.

4.2 Dataset Enrichment

We planned our dataset to be more diverse and to generalize well with existing detectors. This work targets the problem of general video manipulation forensics, extending beyond face-centric detection from ExDDV [2] to encompass spatial, temporal, and audio-level manipulations. Unlike prior approaches that focus primarily on facial artifacts, our scope addresses manipulations applied to arbitrary regions of a video and across multiple modalities.

To support this objective, we curated an explanation-augmented dataset based on **VideoSham** [5], developed by the Adobe Research Team. The Adobe team hired two video engineers to perform six different manipulations using Adobe Effects. The dataset includes the original video, the processed video, a text description of the manipulation, and additional metadata fields. Overall, this dataset setup supports omni-modality analysis and explainable forensic reasoning.

4.2.1 VideoSham Dataset

We selected VideoSham [5] because of its diverse nature. The dataset contains full-scene video alterations affecting spatial layout, temporal consis-

tency, text overlays, and audio tracks, making it suitable for general-purpose forensic analysis.

We categorize manipulations into six distinct types:

1. **Adding an Entity:** A new subject or entity is digitally inserted into the scene, often introducing inconsistencies in scale, motion continuity, or interaction with surrounding objects.
2. **Removing an Entity:** An existing subject is removed, resulting in missing interactions, spatial discontinuities, or residual visual artifacts.
3. **Background / Color Change:** Scene elements such as walls or objects undergo color or texture modification, potentially violating lighting consistency or shadow alignment.
4. **Text Replaced / Text Added:** On-screen text overlays are altered or inserted, frequently causing semantic inconsistencies with spoken dialogue or abrupt visual transitions.
5. **Inter-Frame (Temporal) Manipulation:** Frames are duplicated, dropped, or reordered, producing unnatural motion patterns or repeated visual content.
6. **Audio Replaced / Audio Added:** The audio track is substituted or augmented, resulting in audio-visual desynchronization or semantically implausible sound sources.

4.2.2 Dataset Statistics

The VideoSham dataset contains 400 original and 400 processed videos. For this project, we selected a total of 100 annotated videos from the repository and split them as follows:

- **Training Set:** 70 videos
- **Testing Set:** 30 videos

The distribution of all six manipulation types is approximately balanced across both splits as shown in Fig. ???. The average explanation length is 32.9 words for the training set and 32.5 words for the testing set, encouraging concise yet informative forensic descriptions.

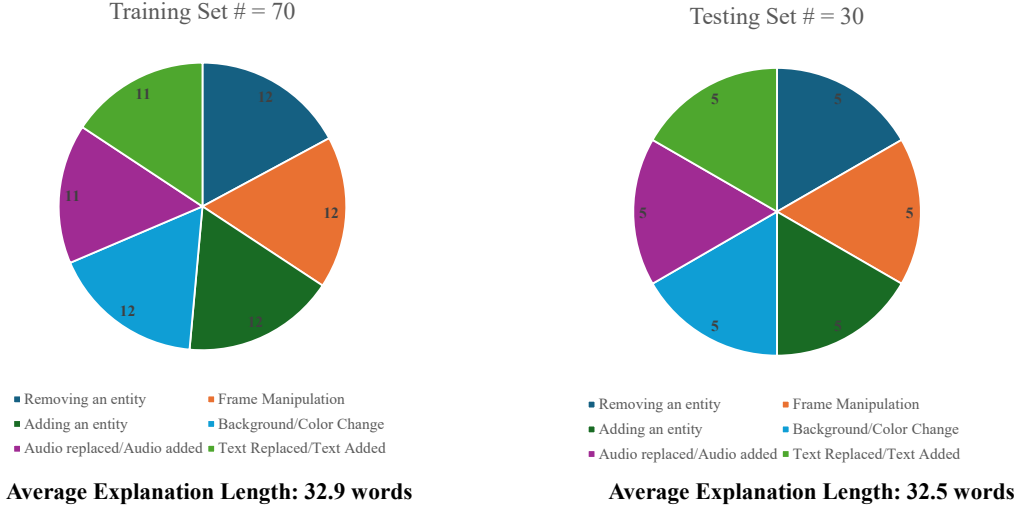


Figure 1: Dataset statistics for the curated VideoSham subset. The figure shows the distribution of the six manipulation types across the training and testing splits, which are approximately balanced.

4.2.3 Annotation and Explainability Pipeline

The manipulation text descriptions provided in VideoSham are not always clear or easy to interpret for someone new to the dataset. So, we worked on enriching the manipulation summaries. This was an iterative process where we generated summaries using the state of the art multimodal large language model Gemini, and refined the summaries based on the feedback from the mentors. We were interested in the following components in the language model response:

- what appears manipulated,
- where or when anomalies occur, and
- supporting visual, temporal, or audio indicators
- summary of description

4.2.4 Prompt Structure

Explanations are generated using a structured forensic prompt that positions the model as a video forensics expert analyzing a single processed

video. The prompt includes:

Task Description:

- A task description specifying forensic analysis objectives and assigning a role to the model

Input Description:

- Processed and unedited video inputs
- task description meta data field
- manipulation description field

Regularization: To ensure trustworthy and evidence-based explanations, we enforce the following constraints:

- Uncertainty Awareness: Conclusions must be phrased cautiously using probabilistic language (e.g., “appears”, “suggests”).
- No Hallucination: Explanations are restricted to entities and events observable in the processed video.
- No Identity Assumptions: Neutral descriptors such as “subject” or “individual” are used unless attributes are visually explicit.
- Single-Video Evidence: The final explanation should be grounded in evidence from the processed video and should not cite or rely on the original footage in the write-up.

4.2.5 Human–Model Refinement Loop

Initial explanations are generated using the multimodal large language model Gemini, conditioned on a structured prompt that includes a task description, video input, and regularization constraints. The model is instructed to produce a forensic explanation covering where or when manipulation occurs, what appears manipulated, and the associated visual, audio, or temporal artifacts, followed by a concise summary.

The generated explanations are then reviewed by human mentors in a human-in-the-loop refinement process. Feedback focuses on common issues such as overly general descriptions, excessive certainty, or failure to incorporate the provided contextual information. Based on this feedback, the prompt formulation and explanation content are revised, and the model is re-invoked to generate an improved explanation.

This iterative revision loop continues until the explanations achieve sufficient forensic clarity, cautious phrasing, and consistency with observable evidence in the video. The refined explanations produced through this process are used as ground-truth annotations for downstream model training and evaluation. Figure 2 illustrates the human-model refinement loop used to iteratively improve explanation quality.

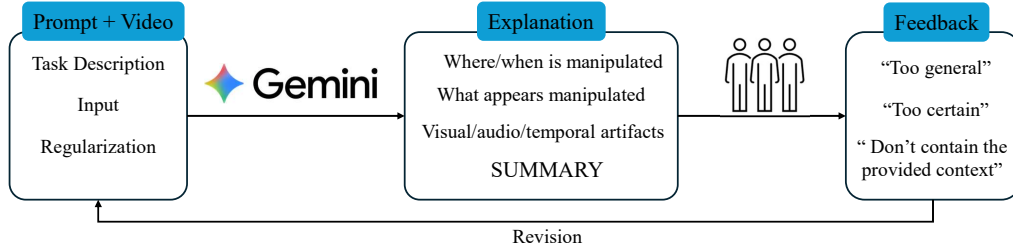


Figure 2: Human-model refinement loop for explanation annotation. A structured prompt and processed video are provided to a multimodal large language model to generate an initial forensic explanation, which is then reviewed by human mentors. Feedback is used to revise the prompt and explanation iteratively until the output achieves sufficient forensic clarity, cautious phrasing, and consistency with observable evidence.

4.3 Model Setup

We employ an open-source omni-modality large language model, Qwen-2.5-Omni-7B [9], to generate and evaluate forensic explanations. The model is capable of jointly processing video, audio, and textual inputs, making it suitable for explainable multimedia forensics involving spatial, temporal, and audio-level manipulations. We evaluate the model under both zero-shot and fine-tuned settings to assess its inherent reasoning ability and its adaptability to the explainable forensics task.

Each input sample consists of a manipulated (processed) video paired with a structured textual prompt. For model evaluation, we use a concise, task-specific prompt that frames the model as a video forensics expert analyzing a single video for signs of digital manipulation. When manipulation is suspected, the model is instructed to produce a single-sentence forensic analysis based solely on the video content. The prompt explicitly requires the explanation to include two components within one sentence: (1) WHERE/WHEN, specifying the spatial region or temporal range in which anomalies are observed, and (2) WHAT APPEARS MANIPULATED, describing the suspected inconsistencies using cautious language. This fixed prompt formulation standardizes the output format across samples and enables direct comparison between zero-shot and fine-tuned outputs.

In the zero-shot setting, Qwen-2.5-Omni-7B generates explanations using only the designed prompt and the manipulated video input, without exposure to task-specific training examples. This setting serves as a baseline to evaluate the model’s inherent ability to reason about multimedia manipulations and generate forensic-style explanations.

To adapt the model to the explainable forensics task, we apply LoRA-based fine-tuning on the curated VideoSham training set. During fine-tuning, the base parameters of Qwen-2.5-Omni-7B are frozen, and only low-rank adaptation layers are updated, enabling parameter-efficient training with limited annotated data. Each training sample consists of a manipulated video and the same task-specific prompt used during evaluation, with the model supervised to generate ground-truth forensic explanations derived from the annotation pipeline.

Fine-tuning is performed using a standard cross-entropy loss over the generated explanation tokens, treating explanation generation as a conditional language modeling task. As illustrated in Fig. 3, the prompt and processed video are encoded by the omni-modality model to produce a textual prediction, which is compared against the ground-truth explanation at the token level. Gradients are backpropagated only through the LoRA parameters, allowing the model to align its multimodal reasoning with human-authored forensic explanations while preserving its general multimodal understanding capabilities.

Fine-tuning is conducted for a limited number of epochs to mitigate overfitting given the dataset size. Specifically, the model is fine-tuned for three epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 1. A cosine learning rate scheduler with a warmup ratio of 0.1

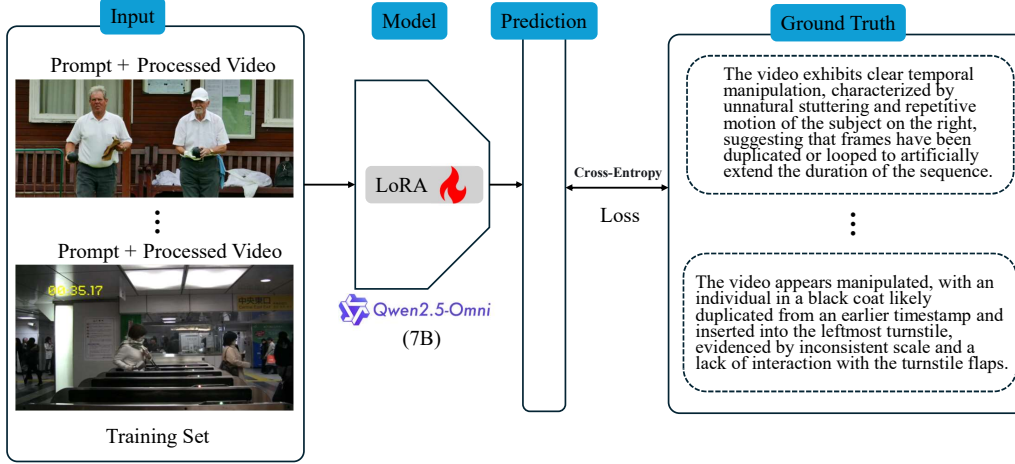


Figure 3: Overview of the omni-modality model fine-tuning pipeline. The model takes a processed video and a task-specific prompt as input, generates a forensic explanation, and is fine-tuned using LoRA with token-level cross-entropy loss against ground-truth explanations.

is applied to stabilize training. During fine-tuning, the vision and audio towers of the omni-modality model are frozen, and only the LoRA parameters are updated. Video inputs are uniformly sampled at 2 frames per second, consistent with the input representation used during evaluation.

4.4 Procedures

The procedures of the project consist of three main stages: dataset preparation, model evaluation, and performance analysis.

Dataset Preparation. Manipulated videos are selected from the curated VideoSham subset, and structured explanations are generated and refined through iterative human-model interaction. Quality control is performed to ensure explanation clarity, consistency, and grounding in observable evidence.

Model Evaluation. The Qwen-2.5-Omni-7B model is first evaluated in a zero-shot setting using only the designed prompt and processed video inputs. The model is then fine-tuned using LoRA adaptation on the training split of the dataset. Both models are evaluated on a held-out test split to assess generalization across manipulation types.

Performance Analysis. Model outputs are evaluated using automatic metrics, including BLEU-1 and semantic cosine similarity, as well as human evaluation of explanation consistency and correctness. Quantitative results are complemented by qualitative analysis highlighting both successful explanation cases and common failure modes, particularly for subtle temporal manipulations.

4.5 Schedule

The project followed a staged execution plan aligned with course milestones, as shown in Table 1. All major phases have been completed at the time of writing.

Phase	Key Activity	Status
Project Planning	Problem definition and literature review	Completed
Dataset Curation	VideoSham-based dataset selection and explanation annotation	Completed
Midterm Review	Midterm report and presentation	Completed
Model Evaluation	Zero-shot and fine-tuned omni-model experiments	Completed
Analysis	Quantitative and qualitative result analysis	Completed
Final Report	Final report and presentation preparation	Completed

Table 1: Project timeline and completion status

4.6 Deliverables

The completed project deliverables include:

- An explainable multimedia forensics dataset derived from VideoSham with six manipulation categories.
- A prompt-based explanation framework that guides omni-modality large language models to produce structured, evidence-based forensic explanations.

- Zero-shot and fine-tuned evaluation results for Qwen-2.5-Omni-7B using automatic and human evaluation metrics.
- A comprehensive final report and presentation summarizing methods, findings, and limitations.

4.7 Limitations and Delimitations

A primary limitation observed in this work is the difficulty of detecting and explaining subtle frame-level temporal manipulations. While human observers can often infer dropped frames or unnatural motion from temporal discontinuities, the evaluated model lacks a robust representation of temporal continuity, resulting in reduced sensitivity to such artifacts.

Additionally, the dataset size and number of manipulation categories, while diverse, remain constrained by annotation effort and course timelines. This work focuses on explanation generation rather than end-to-end manipulation detection accuracy. These delimitations suggest opportunities for future work in scaling datasets and improving temporal reasoning capabilities in omni-modality models.

5 Findings

5.1 Findings Overview

This section summarizes the key empirical findings of the project. We evaluate the ability of an omni-modality large language model to generate forensic explanations for general video manipulations under both zero-shot and fine-tuned settings. Our analysis combines automatic metrics and human evaluation to assess explanation quality across multiple manipulation types.

Overall, the results show that LoRA-based fine-tuning substantially improves the quality, specificity, and consistency of generated forensic explanations compared to the zero-shot baseline. Fine-tuned models produce explanations that more accurately describe where or when anomalies occur and what appears manipulated, while maintaining cautious and forensic-style language. However, performance varies across manipulation categories, with temporal manipulations remaining the most challenging.

5.2 Detailed Findings

Zero-shot Model Evaluation. Figure 4 presents an example of forensic explanation generated by the Qwen-2.5-Omni-7B model in the zero-shot setting. The model correctly suspects that the video has been manipulated; however, the generated explanation deviates from the ground-truth interpretation. Instead of identifying the removal of an individual and the persistence of residual audio, the model attributes the manipulation to abrupt background lighting changes around the 1.5-second mark, suggesting a generic visual alteration.

Quantitatively, the zero-shot model achieves a BLEU-1 score of 0.2326 and a semantic cosine similarity of 0.5766, reflecting limited alignment with the ground-truth explanation. Human evaluation further reports an explanation accuracy of only 16.67%, despite a much higher human baseline accuracy of 83.33%. These results indicate that, without task-specific adaptation, the model tends to rely on superficial visual cues and struggles to jointly reason across visual and audio modalities, leading to incorrect or incomplete forensic explanations.

Fine-tuned Model Evaluation. Figure 5 illustrates an example of forensic explanation generated by the fine-tuned Qwen-2.5-Omni-7B model. The model correctly identifies an audio manipulation event and localizes it to a specific temporal segment of the video. Compared to the ground-truth explanation, the predicted output captures both the nature of the manipulation and the supporting evidence, such as the abrupt replacement of ambient sound and the resulting audio-visual desynchronization.

Quantitatively, the fine-tuned model achieves a BLEU-1 score of 0.2944 and a semantic cosine similarity of 0.6156, indicating improved lexical overlap and semantic alignment with ground-truth explanations. Human evaluation further reports an explanation accuracy of 50%, suggesting that half of the generated explanations are judged to be consistent with the annotated forensic interpretation. Together, these results demonstrate that fine-tuning enables the model to produce explanations that are more specific, grounded, and semantically aligned with human-authored forensic descriptions.

Failure Cases. Despite the overall improvements from fine-tuning, certain manipulation types remain challenging. Figure W presents a representative failure case involving inter-frame temporal manipulation. As indicated by

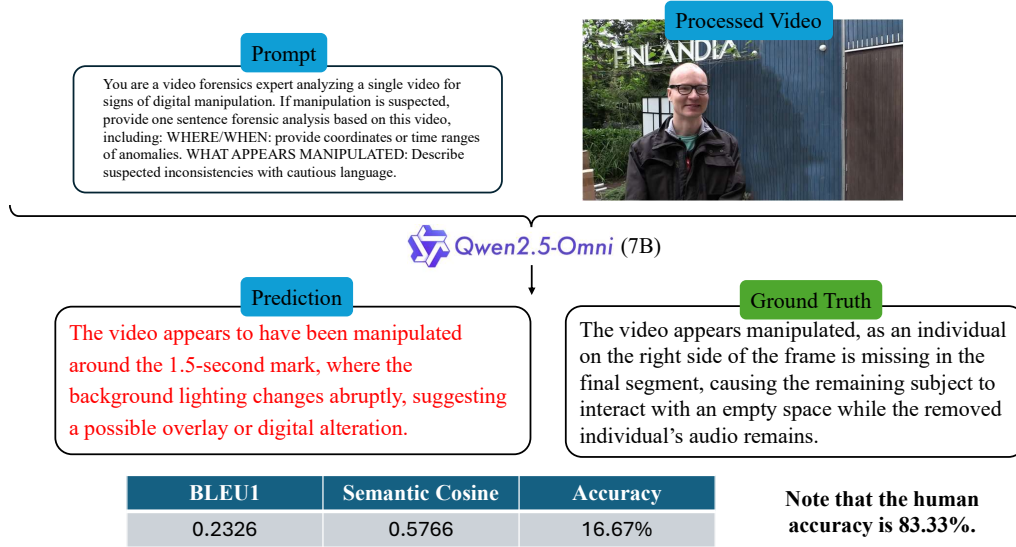


Figure 4: Example of zero-shot model evaluation. Given a processed video and a task-specific forensic prompt, the Qwen-2.5-Omni-7B model generates a single-sentence forensic explanation without task-specific fine-tuning. The generated explanation is compared against the ground-truth annotation, and evaluated using BLEU-1, semantic cosine similarity and human judge.

the ground-truth annotation, the video exhibits persistent jerky motion and temporal discontinuities, suggesting that frames have been systematically dropped, resulting in an unnaturally accelerated playback.

In contrast, the model incorrectly attributes the anomaly to spatial compositing artifacts, hypothesizing that distant background structures have been digitally inserted into the scene. Although the model correctly flags the video as manipulated, it fails to identify the underlying temporal cause of the anomaly. This misattribution highlights a limitation of current omni-modality models in representing temporal continuity, where temporal disruptions are often mistaken for spatial inconsistencies. The example suggests that stronger temporal modeling and motion-aware representations are necessary for reliable explanation of frame-level manipulations.

Zero-shot and Fine-tuned Comparison. Figure Z provides a direct comparison between zero-shot and fine-tuned model behavior on the same processed video. Quantitatively, fine-tuning leads to consistent improvements

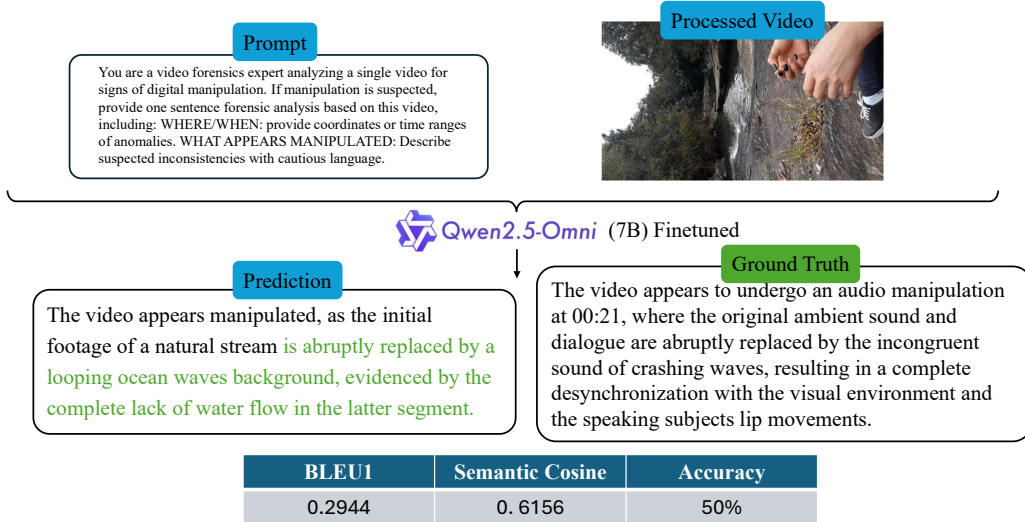


Figure 5: Example of fine-tuned model evaluation. Given a processed video and a task-specific forensic prompt, the fine-tuned Qwen-2.5-Omni-7B model generates a single-sentence forensic explanation, which is compared against the ground-truth annotation. Automatic metrics (BLEU-1 and semantic cosine similarity) and human evaluation accuracy are used to assess quality.

across all evaluation metrics, including BLEU-1, semantic cosine similarity, and human explanation accuracy. In particular, explanation accuracy increases substantially, indicating that fine-tuned explanations are more frequently judged as consistent with the ground-truth forensic interpretation.

Qualitatively, the comparison reveals a clear shift in reasoning behavior. In the zero-shot setting, the model focuses on generic visual cues such as lighting and shadow inconsistencies, resulting in an explanation that does not correspond to the actual manipulation. In contrast, the fine-tuned model correctly identifies the semantic inconsistency introduced by text replacement and identity substitution, aligning closely with the ground-truth explanation. This example illustrates that fine-tuning improves not only metric scores but also the model’s ability to attend to semantically meaningful forensic evidence, producing explanations that are more precise, relevant, and interpretable.



Ground Truth

The footage displays persistent jerky motion and temporal discontinuities throughout, strongly suggesting that frames have been systematically dropped, resulting in an unnatural, accelerated playback speed.

Prediction

The video appears manipulated, as the distant buildings visible in the second segment **seem to have been digitally composited into the background**, evidenced by their lack of interaction with the water and the subject failure to react to their presence.

➤ **Frame manipulation is difficult for AI assistant to detect while easy for human.**

Figure 6: Example of a failure case in fine-tuned model explanation. Although the fine-tuned Qwen-2.5-Omni-7B model correctly flags the video as manipulated, it misattributes a temporal frame-dropping artifact to a spatial compositing inconsistency. This example highlights the difficulty of explaining inter-frame temporal manipulations with current omni-modality models.

6 Issues and Challenges

Several challenges have emerged during the first phase, extending beyond those originally anticipated.

Existing multimodal large language models are not yet strong enough for fine-grained video and audio understanding. Even with the use of Gemini 2.5 Pro for data enrichment, annotation still required significant human screening. Automatically generated descriptions often lacked consistency or precision, creating extra review cycles. We refined our annotation guidelines and ran smaller pilot batches to establish clearer quality thresholds before scaling.

After the government shutdown phase, communication with mentors and access to updated guidance became less immediate. To mitigate this, we have established clearer internal documentation and structured review checkpoints to maintain continuity.

Another key difficulty lies in explanation fidelity. Models may produce rationales that appear convincing but are not grounded in the actual evidence used for classification. To mitigate this, we will enforce consistency between decision logic and generated explanations, and validate outputs using both automated semantic metrics (e.g., BLEU, ROUGE, METEOR, BERTScore)

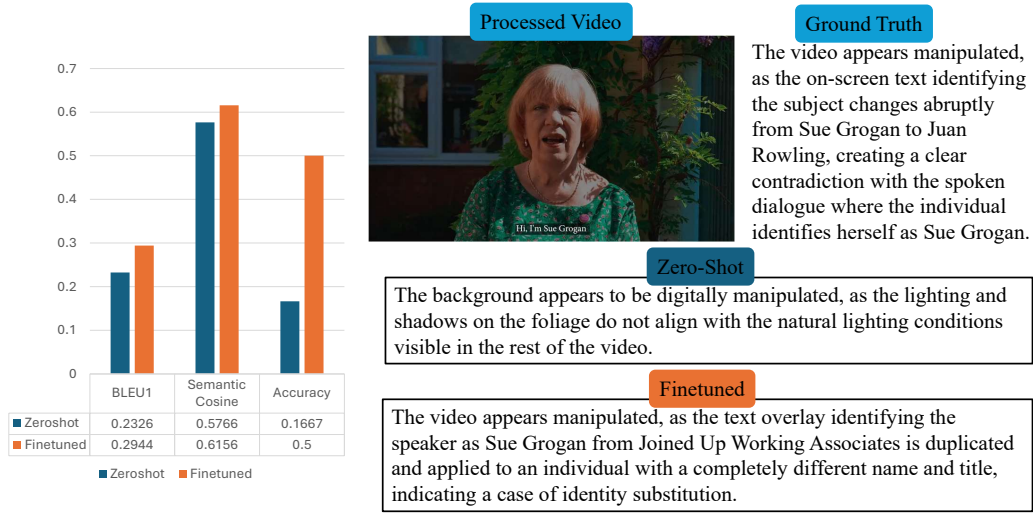


Figure 7: Zero-shot and fine-tuned model comparison on the same processed video. The figure contrasts explanation quality between the zero-shot and fine-tuned Qwen-2.5-Omni-7B models, alongside quantitative metrics including BLEU-1, semantic cosine similarity, and human evaluation accuracy. While the zero-shot model relies on generic visual cues and produces a misaligned explanation, the fine-tuned model correctly identifies the underlying manipulation and generates a more semantically grounded forensic explanation.

and expert review.

Annotation complexity also poses a challenge, as reliable rationales and localization cues require careful guidance. We will pilot small-scale annotation tasks, measure inter-annotator agreement, and refine guidelines before scaling up to ensure high-quality labels.

Additional challenges include the lack of standardized measures for explainability, the difficulty of integrating outputs into analytic workflows, and the need to communicate uncertainty in a clear and usable form. To overcome these, we will define multi-dimensional evaluation protocols that combine semantic quality, localization accuracy, and human utility. User studies will guide the design of interfaces that deliver concise, actionable explanations while minimizing cognitive load. We will also develop calibrated confidence scores, warnings for out-of-scope inputs, and transparent documentation of model limitations.

By anticipating these challenges and building mitigation strategies into each project phase, we aim to deliver explainability methods that are both rigorous and practical.

7 Conclusions and Recommendations

This project investigates general-manipulation explainable multimedia forensics using omni-modality large language models. Moving beyond face-centric deepfake detection, we curate an explanation-augmented dataset based on VideoSham that covers six common video manipulation types, including spatial, temporal, and audio-level alterations. We further evaluate the capability of an open-source omni-modality model, Qwen-2.5-Omni-7B, to generate forensic explanations under both zero-shot and fine-tuned settings.

Experimental results demonstrate that LoRA-based fine-tuning substantially improves explanation quality compared to the zero-shot baseline. Fine-tuned models achieve higher BLEU-1 and semantic cosine similarity scores and receive significantly better human evaluation accuracy. Qualitative analysis shows that fine-tuning helps the model move beyond superficial visual cues and produce explanations that are more semantically grounded and aligned with the underlying manipulation mechanisms. These findings highlight the importance of task-specific adaptation for explainable multimedia forensics.

At the same time, failure case analysis reveals persistent challenges, par-

ticularly for inter-frame temporal manipulations such as frame dropping or duplication. Although the model can often flag videos as manipulated, it may misattribute temporal artifacts to spatial inconsistencies, underscoring limitations in temporal reasoning. Overall, this work provides empirical evidence that omni-modality large language models hold promise for forensic explanation generation, while also emphasizing the need for improved temporal modeling and robust evaluation.

Based on these findings, we recommend that future explainable forensic systems incorporate task-specific fine-tuning and structured prompts to improve explanation reliability. Human evaluation should remain a critical component alongside automatic metrics to ensure that generated explanations are interpretable and actionable for real-world analysts.

8 Future Work

Several directions can be explored to further advance general-manipulation explainable multimedia forensics. First, expanding the dataset in both scale and diversity would improve model robustness. This includes incorporating additional manipulation types, more complex real-world edits, and longer videos with richer temporal dynamics.

Second, future work should focus on improving temporal reasoning capabilities. Incorporating motion-aware representations, explicit temporal modeling, or auxiliary supervision for frame-level continuity could help models better detect and explain inter-frame manipulations that currently pose challenges.

Third, explanation formats could be extended beyond single-sentence descriptions. Multi-sentence explanations, structured reports, or visual grounding mechanisms such as temporal markers and spatial highlights may provide analysts with more actionable insights and facilitate validation.

Finally, broader human-centered evaluation protocols should be explored. In addition to accuracy and semantic similarity, future studies could assess user trust, decision-making efficiency, and the effectiveness of explanations in realistic forensic workflows. These directions would help bridge the gap between technical model performance and practical deployment in high-stakes multimedia forensics applications.

9 References / Bibliography

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10 Team Biographies / Tasking

- **Kavimayil Periyakoravampalayam Komarasamy** is a second-year master’s student in Computer Science, specializing in Data Science and Intelligent Systems. She has 7.5 years of professional experience as a Lead Software Developer and Data Engineer with Accenture, delivering solutions for a leading multinational retailer. She holds an undergraduate degree in Electronics and Communication Engineering. Currently, she is conducting her master’s thesis in Artificial Intelligence, focusing on probabilistic graphical modeling. Her diverse background has enabled her to build expertise across multiple domains and collaborate effectively with cross-functional teams. Adaptable and quick to learn new technologies, she has evolved from programming in C during her undergraduate studies to advanced AI programming in her graduate research.

TASKS: Kavimayil Periyakoravampalayam Komarasamy led the dataset curation and annotation components of the project. She updated the Dataset Curation section to shift the focus from face-centric manipulation detection to a VideoSham based general video manipulation dataset, enabling broader forensic coverage beyond identity-focused deepfakes. Her work included defining and organizing six manipulation categories adding an entity, removing an entity, background or color modification, text replacement or insertion, inter-frame manipulation, and audio replacement ensuring consistent representation of these categories across the dataset.

Kavimayil was also responsible for compiling and reporting dataset statistics, including train test splits, distribution across manipulation types, and average explanation lengths, providing a transparent foundation for benchmarking and evaluation. There was also work to design and refined the Annotation and Explainability Pipeline, defining

structured prompt templates and enforcing regularization rules such as uncertainty-aware language, prohibition of hallucinated content, and avoidance of unsupported identity assumptions. She implemented a human–model refinement loop that combined initial explanation generation using Gemini with iterative human review and correction.

- **Kian Lak** is a second-year master’s student in Computer Science, specializing in Cyber Security. He has 1.5 years of experience as a Software Developer at Cognizant, working closely with clients to develop and deploy solutions for multiple companies. He holds a undergraduate degree in Computer Science from the University of Texas at Dallas. Currently Kian is working on a publication regarding Defending Machine Learning Models against Adversarial Attacks on Images.

TASKS: Kian Lak led the design and implementation of the contrastive retrieval component of the project. He developed and evaluated pipelines for indexing both authentic and manipulated video samples, enabling real–fake side-by-side retrieval to support explainable forensic analysis. The work done included experimenting with embedding representations for video retrieval, tuning similarity search methods, and selecting appropriate distance metrics to retrieve semantically and visually similar samples.

In addition, Kian collaborated closely with Xinpeng Li and Kavimayil Periyakoravampalayam Komarasamy to integrate the retrieval module into the broader explainability framework, ensuring that retrieved examples could be effectively incorporated into vision–language model prompts and downstream analysis. He also contributed to debugging, validation, and qualitative assessment of retrieval outputs.

- **Xinpeng Li** is currently a second-year Ph.D. student in Computer Science at The University of Texas at Dallas, advised by Prof. Yapeng Tian. He holds both a Master’s and Bachelor’s degree in Signal and Information Processing from South China University of Technology, where his work focused on emotion recognition and information engineering. His academic journey has given him a strong foundation in signal processing, machine learning, and multimodal data analysis. Over the past several years, Xinpeng has gained experience across multiple labs and research projects, spanning topics such as multimodal emotion recognition, 3D anomaly detection, and rail detection. These

projects have helped him develop practical skills in deep learning, computer vision, audio–visual modeling, and evaluation of real-world data under noisy and uncertain conditions. His work has led to publications in top venues such as ACM Multimedia, IEEE Transactions on Multimedia, and NeurIPS Datasets and Benchmarks. Currently, his doctoral research focuses on multimodal social interaction understanding and large language models for vision–audio–speech inputs, which has strengthened his expertise in building AI systems that can extract and explain patterns from complex data. His long-term goal is to advance AI techniques that can robustly understand multimodal signals and support security and human-centered applications.

TASKS: Xinpeng Li led the model development, training, and evaluation components of the project. He was responsible for updating the Model and Training Setup. This work involved configuring training parameters such as learning rate, batch size, number of epochs, and frame sampling rate (FPS), as well as determining which model components should remain frozen during adaptation.

Xinpeng also designed and implemented the Evaluation and Results framework. He defined quantitative evaluation metrics, including BLEU-1, semantic cosine similarity, and human accuracy, and conducted systematic comparisons between zero-shot and fine-tuned models. In addition to quantitative analysis, he curated qualitative examples and analyzed failure cases to identify strengths and limitations of the models, particularly in handling subtle spatial, temporal, and audio-level manipulations.

A Acronyms and Terms

A.1 Acronyms

- **GPU (Graphics Processing Unit):** A processor specialized for parallel computation, widely used to accelerate deep learning training and inference.
- **VLM (Vision-Language Model):** A multimodal model that jointly processes visual (image or video) and textual inputs, enabling tasks such as explanation generation and reasoning.

- **Omni-LLM (Omni Large Language Model):** A multimodal model that jointly processes visual, audio, and textual inputs, enabling unified reasoning and explanation generation across modalities.
- **BLEU (Bilingual Evaluation Understudy):** An n-gram overlap-based metric commonly used to measure similarity between generated text and reference text.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** A family of evaluation metrics for text generation and summarization, focusing on n-gram overlap and sequence matching.
- **BERTScore:** A semantic similarity metric that leverages contextual embeddings from pretrained language models (e.g., BERT) to compare generated and reference text beyond surface word overlap.
- **METEOR (Metric for Evaluation of Translation with Explicit Ordering):** A text evaluation metric that considers exact matches, stems, synonyms, and word order, and is often more correlated with human judgment than BLEU.

A.2 Keywords

1. Deepfake Detection
2. Explainability
3. Multimedia Forensics
4. Graphics Processing Unit (GPU)
5. Vision-Language Models (VLMs)
6. Bilingual Evaluation Understudy (BLEU)
7. Recall-Oriented Understudy for Gisting Evaluation (ROUGE)
8. Metric for Evaluation of Translation with Explicit Ordering (METEOR)
9. BERTScore
10. Dataset Enrichment

11. Contrastive Retrieval
12. Multimodal Learning

A.3 Terminology

1. **Deepfake Detection:** The task of identifying whether digital media, such as images, audio, or video, has been artificially generated or manipulated using deep learning techniques.
2. **Explainability:** The ability of an AI system to make its decision-making process transparent and understandable by providing human-interpretable justifications for its outputs.
3. **Multimedia Forensics:** The field of analyzing digital media, including images, audio, and video, to detect, verify, and interpret signs of manipulation or authenticity in security, legal, and investigative contexts.
4. **GPU (Graphics Processing Unit):** A processor optimized for parallel computation, widely used to accelerate training and inference in deep learning systems.
5. **VLM (Vision-Language Model):** A multimodal model that integrates visual and textual inputs to support tasks such as explanation generation and cross-modal reasoning.
6. **BLEU (Bilingual Evaluation Understudy):** An n-gram-based metric for evaluating the similarity between generated text and reference text.
7. **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** A set of evaluation metrics for text generation that emphasize recall-oriented n-gram overlap and sequence matching.
8. **METEOR (Metric for Evaluation of Translation with Explicit Ordering):** A metric that accounts for exact matches, stems, synonyms, and word order, often showing higher correlation with human judgment than BLEU.

9. **BERTScore:** A semantic similarity metric that compares generated and reference text using contextual embeddings from pretrained language models.
10. **Contrastive Retrieval:** A retrieval strategy that leverages both authentic and manipulated samples as side-by-side references to highlight distinguishing features relevant to forensic analysis.
11. **Dataset Enrichment:** The process of expanding a dataset by adding underrepresented manipulation types, modalities, or annotations to improve coverage and robustness.
12. **Multimodal Learning:** A machine learning paradigm that integrates information from multiple modalities, such as text, images, audio, and video, to enable richer and more accurate representations than single-modality approaches.